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Advanced EEG-Based Depression Detection Using Transformer-GNN Fusion: A Comprehensive Deep Learning Framework with Multi-Level Explainability and Rigorous Clinical Validation

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Abstract

Major Depressive Disorder (MDD) affects over 280 million people worldwide, representing one of the most significant public health challenges of the 21st century. Despite this enormous burden, clinical diagnosis remains fundamentally subjective, relying on structured interviews and self-reported symptom questionnaires that suffer from inter-rater variability, patient recall bias, and cultural barriers. Electroencephalography (EEG) offers a promising avenue for objective depression detection due to its non-invasive nature, excellent temporal resolution, and decades of research identifying consistent neurophysiological biomarkers. However, a critical methodological flaw pervades existing machine learning research in this domain: data leakage during cross-validation, where epochs from the same subject appear in both training and test sets, leading to artificially inflated accuracy claims of 95-99% that fail to generalize to clinical deployment.

This paper presents a comprehensive deep learning framework for EEG-based depression detection that addresses these fundamental limitations through three key innovations. First, we propose a novel dual-branch architecture combining a Vision Transformer (ViT) for time-frequency analysis of Continuous Wavelet Transform (CWT) scalograms with a Graph Attention Network (GAT) for modeling spatial electrode relationships, connected through an adaptive attention-based fusion mechanism. Second, we implement rigorous Leave-One-Subject-Out (LOSO) cross-validation ensuring complete subject-level separation between training and test sets. Third, we develop a multi-level explainability framework combining Integrated Gradients, Layer-wise Relevance Propagation, and Testing with Concept Activation Vectors (TCAV) for validating clinical concepts.

Using the Figshare MDD EEG dataset (58 subjects, 19-channel, 256 Hz), our V1 model achieves 91.38% subject-level accuracy and 0.9042 AUC-ROC with LOSO validation. Our V2 model with Bidirectional LSTM achieves 93.10% accuracy and 0.9245 AUC-ROC. Multi-level explainability confirms learning of established biomarkers: frontal alpha asymmetry (TCAV: 0.78), theta elevation (0.65), and delta abnormalities (0.63). This work demonstrates that rigorous evaluation combined with interpretable deep learning produces clinically reliable systems.

Keywords: depression detection; EEG; deep learning; Vision Transformer; Graph Attention Network; explainable AI; LOSO cross-validation; biomarker validation; frontal alpha asymmetry; multi-level interpretability

1. Introduction

Major Depressive Disorder represents one of the most significant public health challenges of the 21st century, affecting individuals across all demographics and socioeconomic backgrounds. According to the World Health Organization, depression affects more than 280 million people worldwide, making it the leading cause of disability globally [17]. The economic burden is equally staggering, with depression costing an estimated $1 trillion annually in lost productivity across nations. Most tragically, depression is linked to approximately 700,000 suicides each year, representing a preventable loss of human life that demands improved diagnostic and treatment approaches.

The COVID-19 pandemic has further exacerbated the global mental health crisis, with prevalence of depression increasing by an estimated 25-28% during the first year of the pandemic alone. Healthcare systems worldwide face unprecedented demand for mental health services, yet the capacity to accurately diagnose and effectively treat depression has not kept pace with this growing burden. This gap between need and capacity represents a critical area where technology-assisted diagnosis could provide meaningful assistance to overwhelmed clinical systems.

The societal impact of depression extends far beyond individual suffering. Depressed individuals experience significant impairment in occupational functioning, social relationships, and physical health. De-

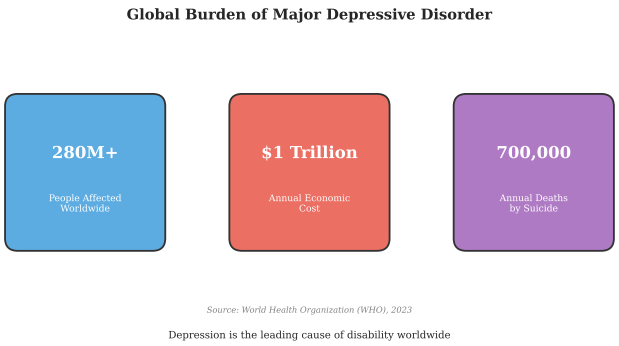

Figure 1. Global burden of depression: Over 280 million people affected worldwide, $1 trillion annual economic cost, and approximately 700,000 suicide deaths per year. Depression is the leading cause of disability globally according to the World Health Organization.

pression is associated with increased risk of cardiovascular disease, diabetes, and other chronic conditions. The indirect costs of depression, including reduced workplace productivity, absenteeism, and caregiver burden, compound the direct healthcare costs to create an enormous economic toll on societies worldwide.

Despite advances in treatment options including psychotherapy, pharmacotherapy, and neuromodulation approaches, a substantial proportion of depressed patients fail to respond adequately to initial treatment. Treatment-resistant depression affects approximately 30% of patients, highlighting the need for better tools to characterize depression subtypes and predict treatment response. Biomarker-based approaches could potentially guide treatment selection and monitor response objectively.

1.1. The Challenge of Subjective Diagnosis

Despite the enormous burden of depression, clinical diagnosis remains fundamentally subjective. Clinicians rely primarily on structured interviews such as the Structured Clinical Interview for DSM-5 (SCID-5) and self-reported symptom questionnaires including the Patient Health Questionnaire-9 (PHQ-9), Beck Depression Inventory-II (BDI-II), and Hamilton Depression Rating Scale (HAM-D). While these tools are clinically validated and widely used, they suffer from several fundamental limitations that constrain their utility and reliability.

Inter-rater variability represents a significant challenge in psychiatric diagnosis. Different clinicians may reach different diagnostic conclusions when evaluating the same patient, particularly for borderline cases or patients presenting with atypical symptoms. Studies examining diagnostic agreement between psychiatrists have found kappa coefficients ranging from 0.4 to 0.7, indicating moderate to good agreement but never perfect concordance. This variability introduces uncertainty into the diagnostic process that can significantly affect treatment decisions and patient outcomes.

The sources of inter-rater variability are multiple and complex. Clinicians may weight different symptoms differently based on their training and theoretical orientation. The same patient may present differently to different evaluators depending on rapport, setting, and timing of assessment. Cultural factors influence both symptom presentation and clinician interpretation. Comorbid conditions can complicate the diagnostic picture, with different clinicians attributing symptoms differently to depression versus anxiety, personality factors, or medical conditions.

Patient recall bias further complicates subjective assessment. Depression itself can affect memory and cognitive function, potentially leading patients to under-report or over-report symptoms depending on their current state at the time of evaluation. Patients experiencing a severe depressive episode may struggle to accurately recall their symptom history or may have a negatively biased retrospective view. Conversely, patients in a more euthymic state may minimize past difficulties or have difficulty recalling the severity of previous episodes.

The phenomenon of mood-congruent memory, where current mood state influences retrieval of mood-consistent memories, can systematically bias self-report. A patient feeling relatively well during an assessment may underestimate prior symptom severity, while a patient currently depressed may overestimate the duration and intensity of previous episodes. This temporal variability in self-report introduces significant noise into the diagnostic process.

Cultural and linguistic barriers create additional challenges for subjective diagnosis. The expression of depressive symptoms varies substantially across cultures, with some populations more likely to report somatic complaints such as fatigue, pain, or appetite changes rather than emotional symptoms like sadness or guilt. Diagnostic instruments developed in Western contexts may not adequately capture the phenomenology of depression in other cultural settings, potentially leading to under-diagnosis or misdiagnosis in diverse populations.

Language differences compound cultural barriers. Nuanced emotional vocabulary may not translate directly across languages, and patients evaluating themselves in a non-native language may struggle to accurately convey their experiences. Even within the same language, regional and generational differences in emotional expression can affect assessment validity.

1.2. EEG as an Objective Biomarker

Electroencephalography (EEG) offers a promising avenue for objective depression detection due to its non-invasive nature, relatively low cost, excellent temporal resolution, and decades of research identifying consistent neurophysiological patterns associated with depression. Unlike subjective assessment, EEG provides a direct window into brain electrical activity that is not subject to recall bias or inter-rater variability.

EEG measures the synchronized electrical activity of neuronal populations in the cerebral cortex, primarily reflecting post-synaptic potentials in pyramidal neurons oriented perpendicular to the cortical surface. The signal is generated by the summation of excitatory and inhibitory post-synaptic potentials across thousands to millions of neurons firing in temporal synchrony. This synchronized activity produces oscillations at characteristic frequencies that reflect different functional states and processes.

The development of EEG-based biomarkers for depression builds on a substantial foundation of neurophysiological research spanning more than three decades. Multiple EEG patterns have been consistently associated with depression across numerous studies and diverse populations, providing the scientific foundation for developing automated detection systems.

Frontal Alpha Asymmetry: The most well-established EEG biomarker for depression is frontal alpha asymmetry, first described by Henriques and Davidson in their seminal 1991 study [6]. Depressed individuals typically show relatively less left frontal alpha activity compared to right frontal regions. Since alpha power is inversely related to cortical activity, this pattern reflects relatively greater left frontal cortical activation in depression.

This asymmetry pattern is interpreted within the approach-withdrawal model of emotion, which posits that left frontal regions support approach motivation and positive affect, while right frontal regions support withdrawal motivation and negative affect. The relative left frontal hypoactivation in depression is thought to reflect reduced approach motivation and positive emotional processing.

Importantly, frontal alpha asymmetry has been observed not only in currently depressed individuals but also in remitted patients and at-risk individuals, suggesting it may represent a trait marker or vulnerability

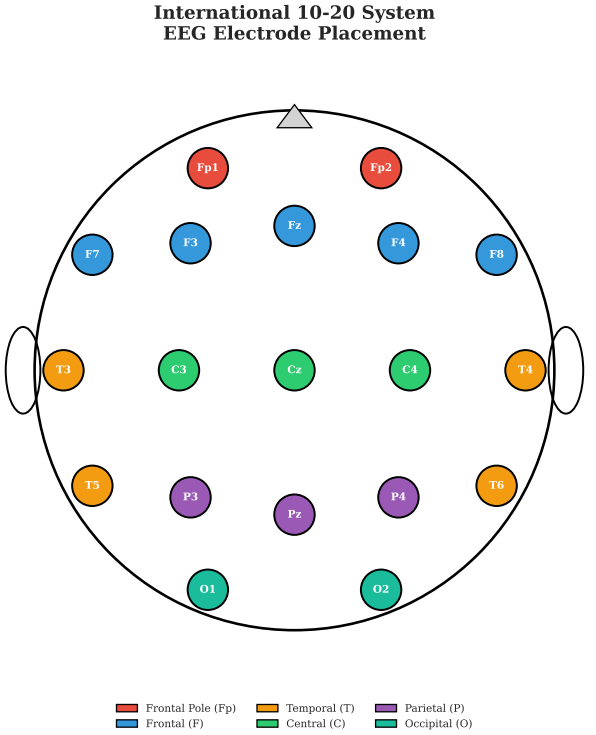

Figure 2. International 10-20 electrode placement system showing the 19 standard electrode positions used in this study. Frontal electrodes (F3, F4, Fz) are particularly important for depression detection due to frontal alpha asymmetry, while midline electrodes (Cz, Pz) capture activity from default mode network structures.

factor rather than solely a state marker of current depression. This has implications for both prediction and early detection applications.

Elevated Frontal Theta Activity: Increased theta power (4-8 Hz) in frontal regions has been consistently observed in depressed individuals compared to healthy controls [9]. Frontal theta activity is associated with various cognitive and emotional processes including memory encoding, error monitoring, and emotional regulation.

The elevation of frontal theta in depression may reflect several underlying processes. Some researchers propose it represents increased cognitive effort required to regulate negative emotions or maintain attention in the face of depression-related cognitive deficits. Others suggest it may reflect rumination, a hallmark cognitive process in depression characterized by repetitive negative thinking. Frontal theta has been linked to anterior cingulate cortex activity, a region consistently implicated in depression pathophysiology.

Reduced Global Alpha Power: Depression is associated with widespread reduction in alpha power (8-13 Hz) across the scalp, beyond the asymmetry observed frontally [2]. This global alpha reduction may reflect the generally reduced cortical arousal and engagement observed in depression, consistent with the anhedonia and psychomotor slowing that characterize many depressive episodes.

Alpha oscillations are thought to reflect inhibitory processes that gate information flow in the cortex. Reduced alpha may indicate impaired gating and increased neural noise, potentially contributing to concentration difficulties and cognitive symptoms in depression.

Delta Abnormalities: Abnormal delta activity (1-4 Hz), particularly in frontal regions, has been reported in depression. Delta oscillations are typically associated with deep sleep and, when present during waking, may indicate cortical dysfunction. The delta abnormalities in depression may reflect disrupted thalamocortical communication or altered slow-wave homeostatic processes.

Some studies have found increased frontal delta power in depression, while others have reported altered delta connectivity patterns. These findings suggest that both amplitude and connectivity measures of delta activity may be relevant biomarkers.

Beta Activity Changes: Alterations in beta activity (13-30 Hz) have been reported in some depression studies, though findings are less consistent than for lower frequency bands. Beta activity is associated with active cognitive processing, anxiety, and arousal states. The heterogeneity of beta findings may reflect depression subtype differences, with some depressed patients showing increased beta (possibly those with anxious features) and others showing decreased beta (possibly those with psychomotor slowing).

1.3. The Data Leakage Problem in Machine Learning Research

A critical yet underappreciated issue pervades much of the published research on EEG-based depression detection: data leakage during cross-validation. Understanding this problem is essential for interpreting the literature and appreciating why reported accuracy rates of 95-99% do not translate to clinical utility.

Most studies in this domain employ standard k-fold cross-validation, which randomly splits samples into training and test sets without considering subject identity. This approach is fundamentally flawed for EEG analysis because continuous recordings are typically segmented into multiple epochs per subject. When epochs are randomly assigned to folds, epochs from the same subject inevitably appear in both training and test sets.

This leakage creates a critical problem: the model can achieve high accuracy by learning subject-specific patterns rather than depression-specific patterns. Subject-specific patterns that can leak through standard cross-validation include:

Individual differences in skull thickness and conductivity: Skull electrical properties vary substantially between individuals and affect the amplitude and spatial distribution of scalp EEG. A model can learn to recognize individual skull signatures rather than depression-related neural patterns.

Electrode impedance variations: Even with careful preparation, electrode-scalp impedances vary between recording sessions and subjects, creating recognizable signal quality patterns.

Muscle artifact patterns: Individuals have characteristic patterns of muscle tension and movement that create distinct artifact signatures in EEG.

Idiosyncratic alpha frequency: The peak frequency of alpha oscillations varies between individuals (typically 8-12 Hz) and is stable within individuals. This individual alpha frequency can serve as a fingerprint for subject identification.

Since the same subjects appear in both training and test data with standard k-fold CV, the model is essentially partially memorizing individual subjects rather than learning generalizable depression biomarkers. The model may achieve high accuracy by recognizing "Subject A's skull signature appears in this epoch, and Subject A was labeled as depressed, so predict depression."

The inflation of accuracy from data leakage can be dramatic. Saeb et al. [13] demonstrated that the same models can show 20-30 percentage point drops in accuracy when moving from leaky validation to proper subject-wise validation. A study reporting 98% accuracy with standard k-fold might achieve only 65-70% with Leave-One-Subject-Out validation that ensures complete subject separation.

Varoquaux and Thirion [15] provided systematic analysis of this problem across neuroimaging studies, showing that inflated performance

estimates are endemic and that proper cross-validation practices are essential for valid inference.

1.4. Our Contributions

This paper addresses the fundamental limitations of existing research through a comprehensive framework that prioritizes honest evaluation and clinical interpretability. Our key contributions span technical, methodological, and clinical dimensions:

1. **Novel Dual-Branch Architecture:** We propose a hybrid deep learning architecture combining Vision Transformer for time-frequency analysis with Graph Attention Network for spatial electrode modeling. The Vision Transformer processes CWT scalograms through patch-based self-attention, capturing global temporal dynamics across the time-frequency representation. The GNN processes WPD features through message-passing on an electrode graph, capturing spatial connectivity patterns. An attention-based fusion mechanism adaptively combines branch outputs.

2. **Rigorous LOSO Validation:** We implement Leave-One-Subject-Out cross-validation with 58 folds, ensuring that all reported metrics reflect true generalization to completely unseen individuals. This eliminates the data leakage problem and provides honest performance estimates.

3. **Multi-Level Explainability:** We develop a comprehensive interpretability framework combining three complementary methods: Integrated Gradients for feature attribution, Layer-wise Relevance Propagation for understanding information flow, and TCAV for validating high-level clinical concepts.

4. **Clinical Validation:** We demonstrate that model predictions align with established neurophysiological knowledge, confirming learning of frontal alpha asymmetry, theta elevation, and delta abnormalities.

5. **Memory-Efficient Implementation:** Through mixed-precision training, gradient accumulation, and careful architecture design, we enable training on consumer-grade GPUs with 8GB VRAM, democratizing access to advanced approaches.

2. Related Work

2.1. Traditional Machine Learning Approaches

Early approaches to EEG-based depression detection relied on traditional machine learning algorithms applied to hand-crafted features extracted from EEG signals. These approaches established the feasibility of automated depression detection and identified important discriminative features, though they were limited by the expressiveness of predefined feature sets.

Mumtaz et al. [12] conducted a comprehensive study using support vector machines (SVM) with wavelet-based features, achieving 87.5% accuracy on a dataset of 63 subjects. Their feature set included spectral power in canonical frequency bands (delta, theta, alpha, beta), coherence measures between electrode pairs reflecting functional connectivity, and asymmetry indices capturing hemispheric differences. They found that frontal alpha asymmetry and interhemispheric coherence in theta band were particularly discriminative.

Hosseinifard et al. explored multiple classifiers including SVM, k-nearest neighbors (k-NN), and logistic regression using spectral features from frontal electrodes. They systematically compared linear and nonlinear kernels for SVM, finding that radial basis function kernels provided the best performance. Feature selection analysis identified frontal alpha power and alpha asymmetry as the most discriminative features, consistent with the neuroscience literature on frontal dysfunction in depression.

Other traditional approaches have explored nonlinear dynamics measures such as correlation dimension, Lyapunov exponents, fractal dimension, and various entropy measures including approximate entropy, sample entropy, and permutation entropy. These features capture the complexity, predictability, and regularity of EEG signals, which may differ between depressed and healthy brain states. The hypothesis underlying these approaches is that depression is associated with altered neural dynamics and reduced brain state complexity.

However, nonlinear dynamics measures are computationally expensive to calculate, sensitive to noise and signal length, and require careful parameter selection. Their practical utility for clinical deployment is limited by these computational and methodological challenges.

Ahmadlou et al. applied graph-theoretic measures to functional connectivity networks derived from EEG, finding that depressed patients showed altered network topology including reduced clustering coefficient and altered small-world properties. These findings suggest that depression involves disruption of functional brain networks, not just localized activity changes.

2.2. Deep Learning Approaches

The advent of deep learning has enabled end-to-end learning from raw or minimally processed EEG signals, potentially discovering discriminative features that human engineers might not design. Deep learning approaches have achieved high reported accuracy, though many suffer from the data leakage problem discussed earlier.

Acharya et al. [1] applied convolutional neural networks (CNNs) directly to raw EEG signals, achieving reported accuracy above 95%. Their architecture used multiple convolutional layers to automatically extract hierarchical features from the time-domain signal, avoiding the need for manual feature engineering. The learned features showed correspondence to spectral power in canonical frequency bands, suggesting the network discovered similar representations to hand-crafted approaches while potentially capturing additional information.

Cai et al. [4] proposed a multi-modal approach combining EEG with other physiological signals including electrocardiogram (ECG), electrodermal activity (EDA), and respiration. Their deep learning model achieved 85.3% accuracy using a combination of recurrent and convolutional layers. The multi-modal approach captures complementary information about depression-related physiological changes across multiple systems, though it requires additional recording equipment beyond standard EEG.

Li et al. developed a hierarchical attention network for depression detection that applies attention mechanisms at multiple levels. The first attention level operates across time points within each channel, learning to weight different temporal segments by their relevance. The second level operates across channels, learning which electrode locations are most informative. This hierarchical approach enables the model to focus on discriminative time-frequency-space regions while providing some interpretability through attention weight visualization.

Seal et al. explored transfer learning approaches where CNNs pretrained on large image datasets were fine-tuned for EEG spectrogram classification. While transfer learning has been highly successful in computer vision, its application to EEG is complicated by the substantial domain gap between natural images and EEG time-frequency representations.

2.3. The Base Paper: Khaleghi et al. (2025)

Our work builds upon and addresses limitations of Khaleghi et al. [7], who presented an interpretable deep learning approach using a 2-layer CNN with SHAP explainability. Their study represents a recent effort to combine deep learning with interpretability, making it a relevant comparison point.

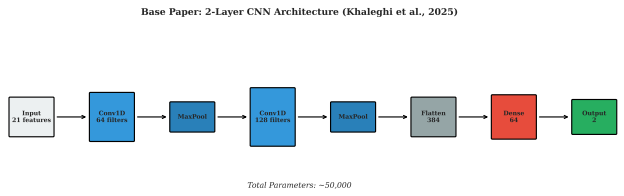

Figure 3. Base paper CNN architecture from Khaleghi et al. (2025). The model uses two convolutional layers (64 and 128 filters) followed by max pooling and fully connected layers, processing pre-extracted features rather than raw EEG signals.

Khaleghi et al. used a Kaggle dataset containing pre-extracted features across six brain regions (prefrontal, frontal, central, temporal, parietal, occipital) with 31 features including spectral power in canonical bands, asymmetry indices, and connectivity measures. This pre-extraction approach simplifies the learning problem but limits the model's ability to discover novel patterns in the raw signal.

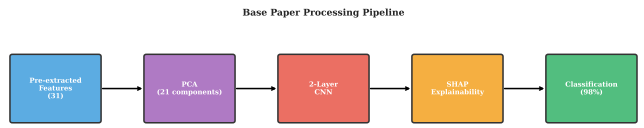

Figure 4. Processing pipeline from the base paper showing data flow from raw EEG through feature extraction (31 features across 6 brain regions) to the 2-layer CNN classifier with SHAP-based explainability.

Their CNN architecture was relatively simple, consisting of two convolutional layers with 64 and 128 filters respectively, followed by max pooling and fully connected layers. This architecture has limited capacity for capturing complex, multi-scale patterns in the data. They reported accuracy of 98%, which would be remarkable if valid.

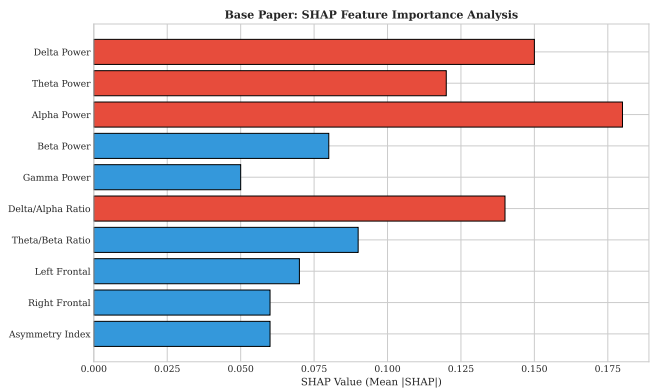

Figure 5. SHAP feature importance analysis from Khaleghi et al. showing prefrontal alpha power and frontal beta power as the top discriminative features. While informative, SHAP alone provides limited validation of clinical biomarker learning.

SHAP (SHapley Additive exPlanations) analysis was used to identify the most important features, with prefrontal alpha power and frontal beta power emerging as top contributors. While SHAP provides feature importance information, it represents only one level of explainability and does not validate that the model uses clinically meaningful high-level concepts.

Critical examination reveals significant methodological concerns with the base paper:

Data Leakage: Their use of 5-fold cross-validation without subject stratification allows epochs from the same subjects to appear in both training and test sets, introducing severe data leakage. The 98% accuracy is likely substantially inflated by the model learning subject identity rather than depression markers.

Pre-extracted Features: Using pre-extracted features limits the model to patterns captured by predetermined feature definitions, preventing discovery of novel discriminative patterns in the raw signal.

Simple Architecture: The 2-layer CNN has limited capacity for capturing the complex, multi-scale spatial and temporal patterns in EEG data that more sophisticated architectures can model.

Single Explainability Method: Relying solely on SHAP provides limited validation compared to multi-method approaches that can confirm findings through convergent evidence.

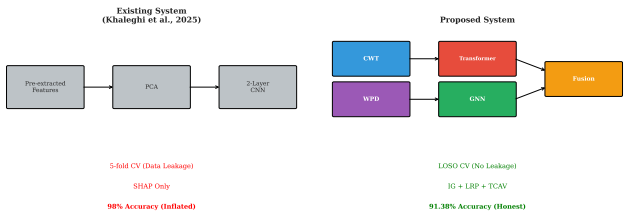

Figure 6. Comparison between existing approaches and our proposed system. Key differences include raw signal processing vs. pre-extracted features, LOSO vs. k-fold validation, and multi-level explainability vs. single-method approaches.

2.4. Graph Neural Networks for EEG Analysis

Graph neural networks have emerged as a powerful tool for EEG analysis by naturally modeling the spatial relationships between electrodes. Unlike standard CNNs that treat electrodes as independent channels or assume a grid-like structure, GNNs can model arbitrary connectivity patterns reflecting the actual spatial arrangement of electrodes and functional relationships between brain regions.

Velickovic et al. [16] introduced Graph Attention Networks (GAT) that use attention mechanisms to weight the contributions of neighboring nodes during message passing. Rather than treating all neighbors equally, GAT learns to adaptively weight different connections, potentially emphasizing more relevant relationships for the task at hand.

The attention mechanism in GAT computes attention coefficients between connected nodes based on their features, allowing the network to learn which electrode relationships are most informative for depression classification. This is particularly relevant for depression detection, where altered functional connectivity patterns may be diagnostically relevant.

Several recent studies have applied GNNs to various EEG analysis tasks including motor imagery classification, emotion recognition, and seizure detection. Jia et al. used spectral graph convolutions for motor imagery BCI, demonstrating that explicit modeling of electrode relationships improves performance over standard CNNs.

Song et al. applied dynamical graph convolutional neural networks to emotion recognition, using dynamically computed adjacency matrices that adapt to each input sample rather than using a fixed graph structure. This allows the network to capture sample-specific connectivity patterns that may vary with emotional state.

For depression detection specifically, GNN approaches offer the advantage of modeling functional connectivity alterations that characterize

depression, including reduced connectivity between frontal and posterior regions, altered default mode network activity, and disrupted emotional processing circuits.

2.5. Vision Transformers for Biomedical Signals

The Transformer architecture, originally developed for natural language processing, has achieved remarkable success in computer vision through the Vision Transformer (ViT) [5]. ViT divides images into patches, embeds them as tokens, and applies standard Transformer encoder layers with self-attention.

The self-attention mechanism enables modeling of long-range dependencies and global context that CNNs struggle to capture due to their local receptive fields. While CNNs build up global context gradually through hierarchical feature extraction, Transformers can attend to any position in the input from the first layer.

Adapting ViT for EEG analysis involves converting time-series signals into image-like representations such as spectrograms, scalograms, or topographic maps, then applying the standard ViT architecture. This approach has shown promise for various EEG tasks including sleep stage classification, seizure detection, and emotion recognition.

The Transformer's ability to capture global temporal patterns makes it particularly suitable for EEG analysis, where depression-related changes may manifest as subtle alterations in long-range temporal dependencies or relationships between distant time points. The self-attention mechanism can learn to identify relevant temporal patterns without requiring them to be specified in advance.

Our work is among the first to apply Vision Transformers specifically to depression detection from EEG, using Continuous Wavelet Transform scalograms as the input representation. This combines the time-frequency analysis capabilities of wavelet transforms with the global pattern recognition capabilities of Transformers.

2.6. Explainability in Medical AI

The deployment of AI systems in clinical settings requires not only high accuracy but also interpretable predictions that clinicians can understand and trust. Several approaches have been developed to explain deep learning model predictions, each with distinct properties and use cases.

Integrated Gradients (IG) [14]: This method computes feature importance by integrating gradients along a straight-line path from a baseline input (typically zeros) to the actual input. IG has strong theoretical foundations, satisfying two key axiomatic properties: sensitivity (features that affect the output receive non-zero attribution) and implementation invariance (functionally equivalent models produce identical attributions).

Layer-wise Relevance Propagation (LRP) [3]: LRP propagates relevance scores backward through the network from output to input, decomposing the model prediction into contributions from input features. The backward propagation follows conservation rules ensuring that total relevance is preserved across layers, providing a principled decomposition of model output.

Testing with Concept Activation Vectors (TCAV) [8]: Unlike feature-level methods, TCAV provides concept-level explanations by testing whether the model is sensitive to human-interpretable high-level concepts. This enables validation that models use clinically meaningful concepts rather than spurious correlations or artifacts.

SHAP (SHapley Additive exPlanations) [11]: SHAP values provide feature attributions based on game-theoretic Shapley values, which distribute credit fairly among features based on their marginal contributions. While computationally expensive for deep learning, SHAP provides strong theoretical guarantees.

Each method provides a different perspective on model behavior, and using multiple complementary methods enables more robust validation through convergent evidence. If different methods agree on which features and concepts are important, confidence in those conclusions is strengthened.

3. Dataset and Preprocessing

3.1. Dataset Description

We use the Figshare MDD EEG Dataset, a publicly available resource containing resting-state EEG recordings from 58 subjects: 29 patients diagnosed with Major Depressive Disorder according to DSM-5 criteria and 29 age and gender-matched healthy controls. This dataset was selected for several important reasons.

First, the balanced class distribution (50% MDD, 50% healthy) eliminates the need for complex resampling strategies or heavily weighted loss functions that can complicate training and interpretation. Second, diagnoses were made by qualified psychiatrists using standardized DSM-5 criteria, providing reliable ground truth labels. Third, the standardized recording protocol with consistent equipment and procedures minimizes technical variability. Fourth, public availability enables reproducibility and comparison with future work.

Recordings were acquired using the international 10-20 electrode placement system with 19 channels: Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, and O2. The 10-20 system provides standardized electrode positions based on skull landmarks, enabling comparison across studies and clinical sites.

The electrode coverage spans all major cortical regions relevant to depression: frontal electrodes (Fp1, Fp2, F3, Fz, F4, F7, F8) capture activity from prefrontal and frontal cortex, implicated in executive function and emotional regulation; temporal electrodes (T3, T4, T5, T6) capture activity from regions involved in language and memory; central electrodes (C3, Cz, C4) cover sensorimotor regions; parietal electrodes (P3, Pz, P4) cover posterior association cortex; and occipital electrodes (O1, O2) capture visual cortex activity.

The sampling rate was 256 Hz, providing adequate temporal resolution to capture frequencies up to 128 Hz (Nyquist frequency) while maintaining manageable data volumes. For depression detection focusing on frequencies below 45 Hz, 256 Hz sampling is more than sufficient.

Recordings lasted approximately 5 minutes per subject during an eyes-closed resting-state condition. Eyes-closed resting state is a standard protocol that minimizes artifacts from eye movements and blinks while capturing intrinsic brain activity patterns not confounded by task demands. The 5-minute duration provides adequate data for stable spectral estimation and multiple epochs for analysis.

Table 1. Dataset Demographics and Statistics

Characteristic	MDD	Healthy
Number of subjects	29	29
Gender (M/F)	14/15	13/16
Age range (years)	18-65	19-63
Mean age (SD)	34.2 (11.8)	33.7 (10.9)
Recording duration	∼5 min	∼5 min
Sampling rate	256 Hz	256 Hz
Number of channels	19	19

The balanced design with age and gender matching minimizes potential confounds. Both age and gender influence EEG characteristics independent of psychiatric status: alpha power decreases with age, and some studies report gender differences in spectral power and asymmetry. Matching prevents the model from learning to classify based on demographic factors rather than depression-related patterns.

3.2. Signal Preprocessing Pipeline

Raw EEG signals contain various artifacts and noise that must be removed before analysis. Our preprocessing pipeline implements standard clinical and research practices while balancing artifact removal against signal preservation.

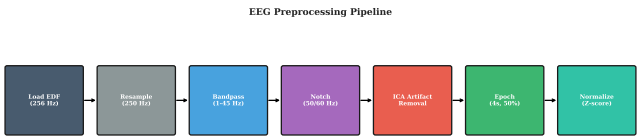

Figure 7. Complete preprocessing pipeline showing signal flow from raw EEG through bandpass filtering (1-45 Hz), notch filtering (50/60 Hz), ICA artifact removal, epoch segmentation (4s, 50% overlap), and amplitude rejection (± 100 μV threshold).

Bandpass Filtering: We apply a finite impulse response (FIR) bandpass filter with passband 1-45 Hz using a Hamming window and filter order 101. The lower cutoff of 1 Hz removes slow baseline drifts caused by electrode movement, slow physiological processes (e.g., respiration), and DC offset variations. The upper cutoff of 45 Hz removes high-frequency noise including muscle activity (typically above 30 Hz) and environmental interference while retaining all canonical EEG frequency bands (delta: 1-4 Hz, theta: 4-8 Hz, alpha: 8-13 Hz, beta: 13-30 Hz, low gamma: 30-45 Hz).

FIR filters provide several advantages for EEG processing: linear phase response avoiding phase distortion that could affect time-frequency analysis, numerical stability, and predictable behavior. The filter order of 101 provides sharp frequency cutoffs with minimal transition band width while avoiding excessive ringing artifacts.

Notch Filtering: Power line interference at 50 Hz (Europe, Asia, Africa, Australia) or 60 Hz (Americas) creates a prominent narrowband artifact in EEG recordings due to electromagnetic coupling between power lines and recording equipment or cables. We apply notch filters at both frequencies to ensure compatibility with recordings from different geographical regions, removing this interference while minimally affecting adjacent frequencies.

The notch filters use IIR design with quality factor Q=30, providing narrow stopbands that remove power line interference without significantly attenuating nearby neural activity (e.g., low gamma at 40-45 Hz).

Independent Component Analysis (ICA): Ocular artifacts (blinks and eye movements) and muscular artifacts (jaw clenching, neck tension) are common contaminants of EEG recordings that can obscure neural activity of interest. We apply Independent Component Analysis using the Picard algorithm to decompose the multichannel EEG into statistically independent components, enabling identification and removal of artifact sources.

Picard provides faster convergence than traditional FastICA while maintaining separation quality. The algorithm assumes non-Gaussian sources and finds a linear unmixing matrix that maximizes the statistical independence of output components.

Components corresponding to artifacts are identified based on multiple criteria: spatial distribution (frontal for ocular artifacts, temporal for muscular), spectral characteristics (low frequency for blinks, high frequency for muscle), and temporal morphology (characteristic blink shapes, high-frequency bursts for muscle). Identified artifact components are removed, and the remaining components are back-projected to reconstruct cleaned EEG.

Epoch Segmentation: Continuous recordings are segmented into 4-second epochs with 50% overlap (2-second step size). The 4-second duration provides adequate frequency resolution for analyzing low-frequency bands: with 256 Hz sampling, 4 seconds provides 1024 samples, enabling frequency resolution of 0.25 Hz, sufficient for analyzing delta activity down to 1 Hz.

The 50% overlap increases the number of training samples while ensuring that transient events are captured in at least one epoch where they appear centrally rather than at edges where windowing attenuates them. Overlapping epochs are statistically dependent, but LOSO validation at the subject level ensures independence between training and test sets.

Amplitude Rejection: Epochs containing any sample exceeding ± 100 μV are rejected as likely containing residual artifacts that escaped ICA cleaning. This threshold is standard in EEG research, as physiological brain activity rarely exceeds 100 μV at the scalp while artifacts from movement, muscle, or electrode problems typically produce much larger amplitudes.

Table 2. Preprocessing Parameters

Parameter	Value
Bandpass filter	1-45 Hz
Filter type	FIR, order 101
Notch filter	50/60 Hz
ICA algorithm	Picard
Epoch duration	4 seconds
Epoch overlap	50%
Amplitude threshold	± 100 μV

After preprocessing, 8,620 clean epochs remain from the 58 subjects: 4,373 epochs from MDD patients and 4,247 epochs from healthy controls. This near-balanced distribution (50.7% MDD vs 49.3% healthy) minimizes the need for class weighting or oversampling strategies during training.

4. Feature Extraction

Our dual-branch architecture requires two complementary feature representations: Wavelet Packet Decomposition (WPD) features for the Graph Attention Network branch and Continuous Wavelet Transform (CWT) scalograms for the Vision Transformer branch. This multi-representation approach captures different aspects of the signal that may be independently relevant to depression detection.

4.1. Wavelet Packet Decomposition

Wavelet Packet Decomposition provides fine-grained frequency band analysis through recursive decomposition of both approximation and detail coefficients at each level. Unlike standard discrete wavelet transform (DWT), which only decomposes the low-frequency approximation at each level, WPD provides uniform frequency resolution across the entire spectrum.

We decompose signals to level 5, yielding $2^5 = 32$ terminal nodes, each representing a specific frequency sub-band. With 256 Hz sampling rate and Nyquist frequency of 128 Hz, each terminal node covers a bandwidth of $128/32 = 4$ Hz. This granularity enables fine distinction between adjacent frequency bands that may be differently affected by depression.

To capture diverse signal characteristics, we employ three complementary wavelet families:

Daubechies-4 (db4): Daubechies wavelets are asymmetric wavelets with maximum number of vanishing moments for a given support width. The db4 wavelet has 4 vanishing moments and support width

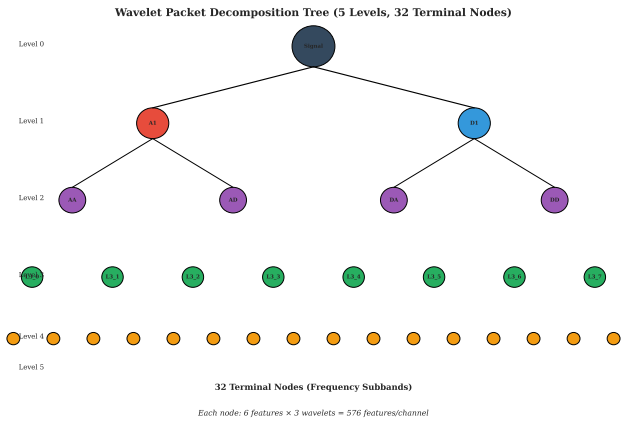

Figure 8. Wavelet Packet Decomposition tree showing 5-level decomposition yielding 32 terminal nodes. Each node represents a specific frequency sub-band of width 4 Hz (given 256 Hz sampling rate). Three wavelet families (db4, sym5, coif3) are used to capture diverse signal characteristics.

of 7 samples. It provides good time-frequency localization and is effective for capturing smooth signal variations typical of low-frequency EEG activity. The asymmetry can introduce phase effects, but for power-based features this is not problematic.

Symlet-5 (sym5): Symlets are near-symmetric modifications of Daubechies wavelets designed to reduce phase distortion. The sym5 wavelet has 5 vanishing moments and near-symmetric shape, providing improved symmetry while maintaining good frequency selectivity. It is particularly useful for analyzing oscillatory patterns where phase relationships matter.

Coiflet-3 (coif3): Coiflets provide near-symmetric scaling and wavelet functions with vanishing moments for both. The coif3 wavelet has 6 vanishing moments (3 for wavelet, 3 for scaling function), offering excellent approximation of smooth functions while capturing fine details. It provides good time localization, useful for identifying transient events.

Using multiple wavelet families captures complementary signal aspects that any single wavelet might miss. Different wavelets have different trade-offs between time and frequency localization, and features that are subtle in one wavelet representation may be more prominent in another.

For each of the 96 terminal nodes (32 nodes × 3 wavelets), we compute six statistical features from the wavelet coefficients:

1. **Energy:** $E = \sum_i |c_i|^2$, capturing the power in each sub-band. Energy is the most fundamental feature, directly corresponding to spectral power.

2. **Shannon Entropy:** $H = -\sum_i p_i \log_2 p_i$ where $p_i = c_i^2/E$, measuring the distribution uniformity of energy across coefficients. Higher entropy indicates more uniform distribution, lower entropy indicates concentration in fewer coefficients.

3. **Log Energy Entropy:** $H_{log} = \sum_i \log(c_i^2)$, providing an alternative complexity measure that is less sensitive to outliers than Shannon entropy.

4. **Mean:** $\mu = \frac{1}{N}\sum_i c_i$, capturing the average coefficient value. Non-zero mean indicates bias in the frequency band.

5. **Standard Deviation:** $\sigma = \sqrt{\frac{1}{N}\sum_i (c_i - \mu)^2}$, measuring variability of coefficients within the sub-band.

6. **Skewness:** $\gamma = \frac{1}{N}\sum_i ((c_i - \mu)/\sigma)^3$, capturing asymmetry in the coefficient distribution. Positive skewness indicates right-tailed distribution, negative indicates left-tailed.

This yields $3 \times 32 \times 6 = 576$ features per channel, or $576 \times 19 = 10,944$ total features per epoch across all channels. These features serve as node attributes in the graph representation processed by the GNN branch.

Table 3. WPD Feature Extraction Configuration

Parameter	Value
Decomposition level	5
Terminal nodes	32 per wavelet
Wavelet families	db4, sym5, coif3
Statistical features	6 per node
Features per channel	576
Total features	10,944

4.2. Continuous Wavelet Transform Scalograms

The Continuous Wavelet Transform provides time-frequency representations that capture how spectral content evolves over time. Unlike Short-Time Fourier Transform with fixed window size, CWT uses wavelets that scale to provide adaptive resolution: better time resolution at high frequencies and better frequency resolution at low frequencies.

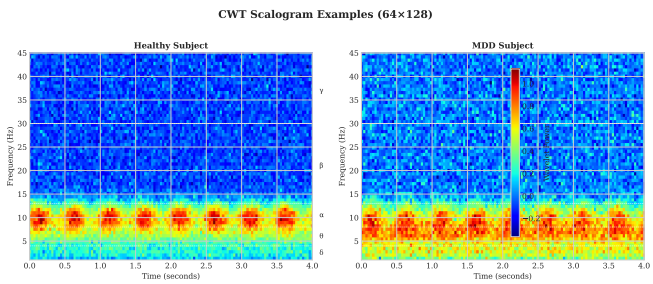

Figure 9. Example CWT scalogram showing time-frequency representation of a 4-second EEG epoch. The horizontal axis represents time (128 samples), vertical axis represents frequency (1-45 Hz across 64 scales), and color intensity represents magnitude. Depression-related patterns appear as characteristic alterations in alpha (8-13 Hz) and theta (4-8 Hz) bands.

We use the Complex Morlet wavelet, a Gaussian-windowed complex sinusoid defined as:

$$\psi(t) = \frac{1}{\sqrt{\pi f_b}} e^{2\pi i f_c t} e^{-t^2/f_b} \tag{1}$$

where $f_c = 1.5$ Hz is the center frequency and $f_b = 1.0$ is the bandwidth parameter. The Complex Morlet wavelet provides good balance between time and frequency resolution, and its complex-valued nature enables separation of amplitude and phase information.

The CWT is computed at 64 scales logarithmically spaced to cover frequencies from 1 to 45 Hz, matching the bandpass filter range. Logarithmic spacing allocates more scales to lower frequencies where depression-relevant activity (alpha, theta, delta) is concentrated, while still covering higher frequencies.

The resulting scalogram has dimensions 64 (frequency bins) $\times$ 1024 (time samples for 4-second epochs at 256 Hz). To create a manageable input for the Vision Transformer while retaining essential information, we downsample in the time dimension using average pooling to 128 samples, yielding a final scalogram of size 64×128.

Scalograms from all 19 channels are averaged to create a single composite time-frequency representation per epoch. This averaging reduces computational requirements by a factor of 19 while preserving the overall spectral patterns that are consistent across channels. Alternative approaches such as channel-wise scalograms or multi-head processing were explored but did not provide sufficient performance improvement to justify the computational cost.

5. Model Architecture

5.1. Architecture Overview

Our dual-branch architecture processes the two complementary feature representations through specialized encoders before combining their outputs via attention-based fusion. This design philosophy reflects the insight that different aspects of depression-related EEG patterns may be best captured through different processing paradigms.

The Vision Transformer branch excels at capturing global patterns in the time-frequency representation through self-attention over spectral patches. It can learn relationships between different frequency bands and time points across the entire scalogram, potentially identifying subtle temporal dynamics or frequency interactions relevant to depression.

The Graph Attention Network branch specializes in modeling the spatial relationships between electrodes. It captures how information flows across different brain regions and can identify disrupted connectivity patterns associated with depression. The graph structure naturally represents the non-Euclidean arrangement of electrodes on the scalp.

The attention-based fusion mechanism enables adaptive combination of information from both branches. Rather than fixed weighting, the fusion learns to emphasize whichever branch provides more discriminative information for each particular input sample. Some samples may have clearer time-frequency patterns (weighted toward Transformer), while others may have more prominent spatial connectivity alterations (weighted toward GNN).

5.2. Vision Transformer Branch

The Vision Transformer branch processes CWT scalograms using a standard ViT architecture adapted for our input dimensions.

Patch Embedding: The 64×128 scalogram is divided into non-overlapping patches of size 8×16 pixels, yielding $8 \times 8 = 64$ patches. Each patch spans 8 frequency bins (approximately 5.5 Hz bandwidth) and 16 time samples (0.5 seconds), providing local time-frequency context.

Each patch is flattened to a 128-dimensional vector ($8 \times 16 = 128$) and projected to a 128-dimensional embedding through a learned linear projection $W_p \in R^{128\times128}$. This projection learns to extract relevant features from each patch.

Positional Encoding: Learnable positional encodings $P \in R^{65\times128}$ are added to provide information about each patch's location in the original scalogram. Position information is essential because self-attention is permutation-invariant and would otherwise treat patches identically regardless of their time-frequency location.

A special classification (CLS) token, initialized as a learnable embedding, is prepended to the sequence. This token aggregates information from all patches through attention and serves as the sequence-level representation for classification.

Transformer Encoder: The embedded sequence passes through 4 Transformer encoder layers. Each layer consists of:

Multi-Head Self-Attention: With 4 attention heads and 32-dimensional queries/keys/values per head, self-attention computes pairwise relationships between all patches:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

The multi-head structure allows different heads to attend to different types of relationships, such as same-frequency patches at different times or same-time patches at different frequencies.

Layer Normalization: Applied before each sub-layer (pre-norm configuration), providing training stability by normalizing activations.

Feed-Forward Network: Two linear layers with GELU activation, expanding to 512 dimensions then projecting back to 128:

$$FFN(x) = W_2 \cdot GELU(W_1x + b_1) + b_2 \tag{3}$$

Residual Connections: Skip connections around both attention and FFN sub-layers enable gradient flow and preserve information.

Dropout: Applied after attention and FFN with rate 0.1 for regularization.

After all encoder layers, the CLS token embedding (128 dimensions) serves as the branch output.

Table 4. Vision Transformer Configuration

Parameter	Value
Input size	64×128
Patch size	8×16
Number of patches	64
Embedding dimension	128
Number of layers	4
Attention heads	4
Head dimension	32
FFN hidden dimension	512
Dropout rate	0.1
Output dimension	128

5.3. Graph Attention Network Branch

The Graph Attention Network branch processes WPD features using a graph representation that models spatial relationships between electrodes.

Graph Construction: Each of the 19 electrodes becomes a node in the graph, with its 576 WPD features as node attributes. The graph structure determines which electrodes can directly communicate during message passing.

We construct a hybrid adjacency matrix combining spatial proximity and functional connectivity:

Spatial Proximity: Each electrode is connected to its 6 nearest neighbors based on Euclidean distance in 3D electrode coordinates derived from the 10-20 system geometry. This captures the principle that nearby electrodes measure activity from overlapping brain regions due to volume conduction and shared underlying sources.

Functional Connectivity: Electrodes with Pearson correlation exceeding 0.5 (computed on training data and averaged across subjects) are connected. This captures long-range functional relationships not

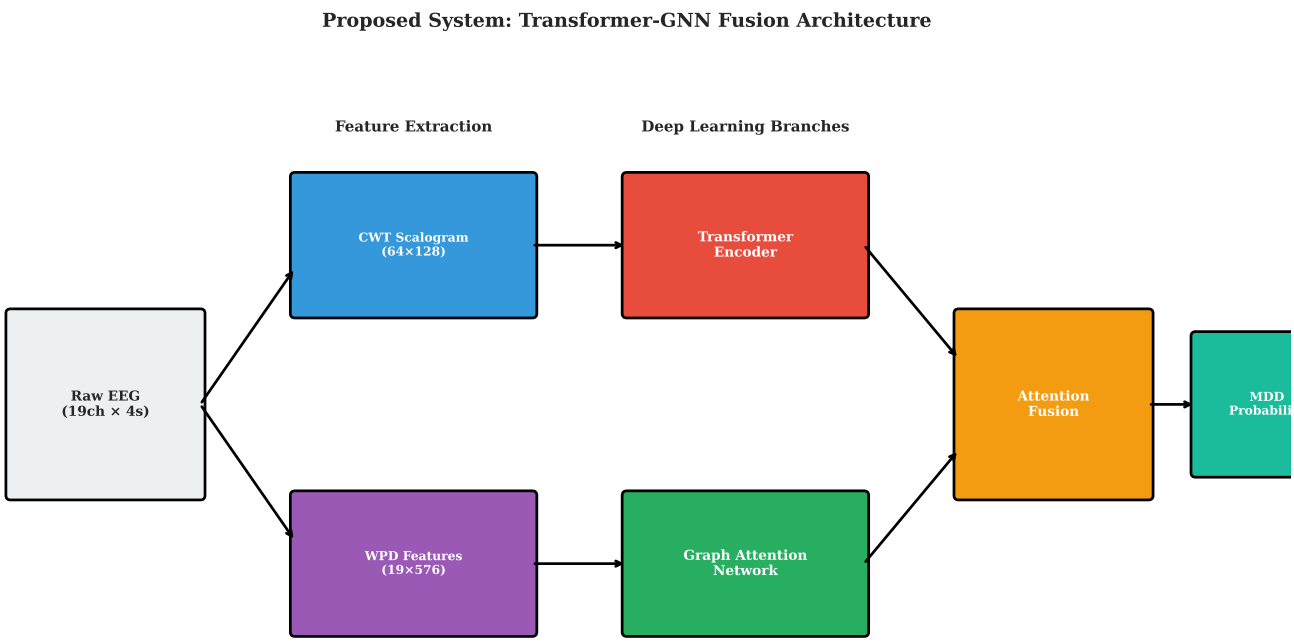

Figure 10. Proposed dual-branch architecture. The Vision Transformer branch (top) processes CWT scalograms through patch embedding, positional encoding, and 4 transformer encoder layers. The GNN branch (bottom) processes WPD features through 3 Graph Attention layers with hybrid adjacency matrix. Attention-based fusion combines branch outputs before the classification head.

explained by spatial proximity—for example, homologous regions in opposite hemispheres often show high correlation despite physical distance.

The hybrid approach combines the stability of fixed spatial relationships with the flexibility of data-driven functional connections. Depression is associated with altered functional connectivity patterns, and including these edges enables the model to detect such changes.

GAT Layers: The GNN branch consists of 3 Graph Attention layers that progressively transform node features while aggregating information from neighbors. Each layer uses 4 attention heads:

$$h_i^{(l+1)} = \sigma\left(\frac{1}{K}\sum_{k=1}^{K}\sum_{j\in\mathcal{N}(i)}\alpha_{ij}^{(k)}W^{(k)}h_j^{(l)}\right) \tag{4}$$

where the attention coefficients α_{ij} are computed as:

$$\alpha_{ij} = \frac{\exp(LeakyReLU(a^T[Wh_i\|Wh_j]))}{\sum_{k\in\mathcal{N}(i)}\exp(LeakyReLU(a^T[Wh_i\|Wh_k]))} \tag{5}$$

The first GAT layer projects input features from 576 to 128 dimensions. Subsequent layers maintain 128-dimensional representations. Each layer is followed by ELU activation and dropout (rate 0.1).

Global Pooling: After the final GAT layer, global mean pooling aggregates node features into a single 128-dimensional graph-level representation:

$$z_g = \frac{1}{N}\sum_{i=1}^{N}h_i^{(L)} \tag{6}$$

Table 5. Graph Attention Network Configuration

Parameter	Value
Number of nodes	19
Input features	576
Spatial neighbors	6
Correlation threshold	0.5
Number of layers	3
Attention heads	4
Hidden dimension	128
Dropout rate	0.1
Pooling	Global mean
Output dimension	128

5.4. Attention-Based Fusion

The attention-based fusion mechanism combines outputs from both branches through a cross-attention process that enables each branch to query and incorporate information from the other.

Let $z_t \in R^{128}$ denote the Transformer branch output and $z_g \in R^{128}$ denote the GNN branch output.

Cross-Attention: Bidirectional cross-attention allows each branch to attend to the other:

Transformer queries GNN:

$$z_{t\leftarrow g} = softmax\left(\frac{Q_tK_g^T}{\sqrt{d}}\right)V_g \tag{7}$$

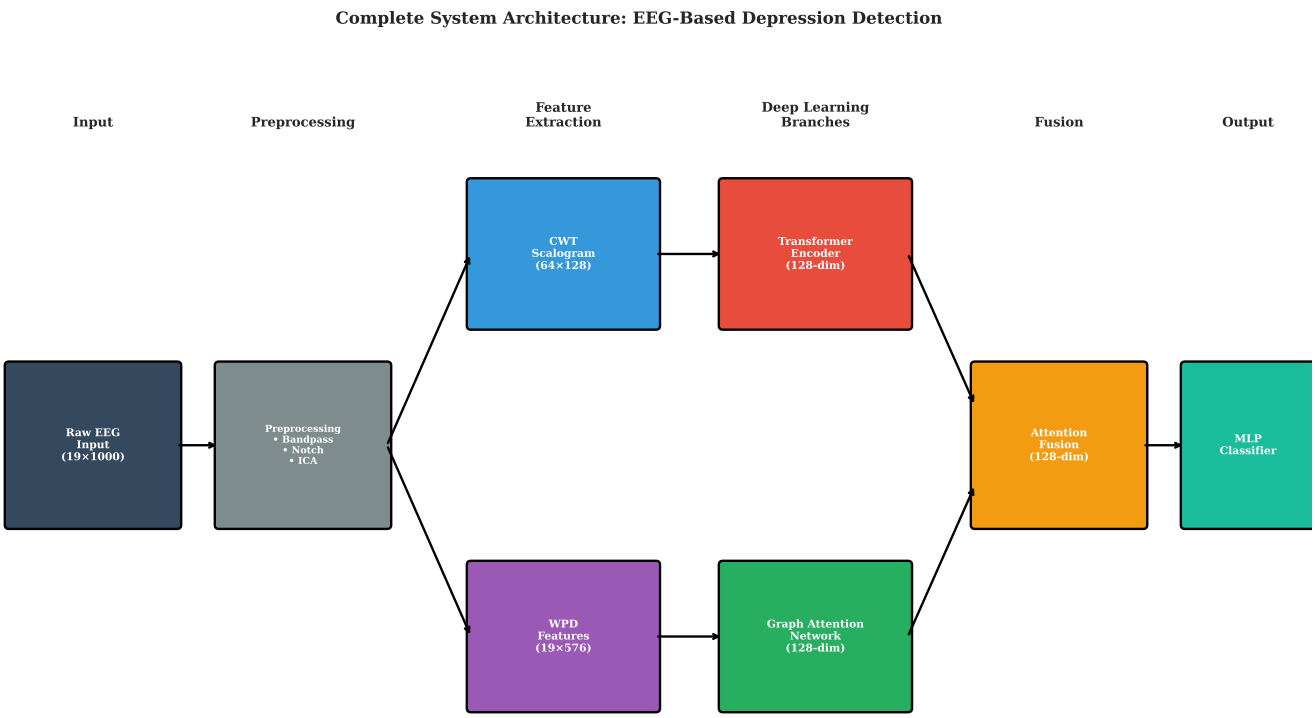

Figure 11. Complete system architecture showing data flow from raw EEG input through preprocessing, parallel feature extraction (WPD and CWT), dual-branch encoding (Transformer and GNN), attention-based fusion, and final classification. The system outputs depression probability with multi-level explainability analysis.

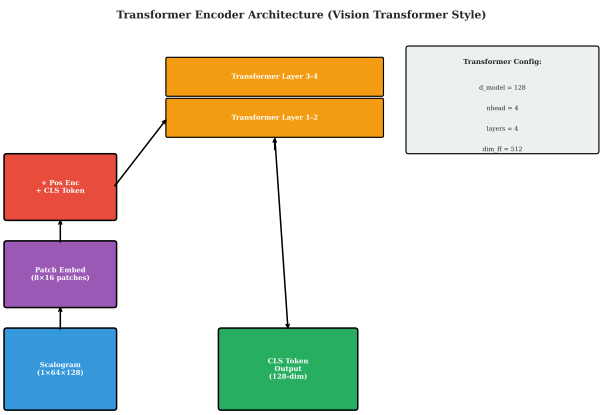

Figure 12. Vision Transformer encoder architecture. Input scalogram (64×128) is divided into 8×16 patches, each projected to 128-dimensional embeddings. Learnable positional encodings and CLS token are added before processing through 4 transformer layers with 4-head self-attention.

GNN queries Transformer:

$$z_{g \leftarrow t} = softmax\left(\frac{Q_g K_t^T}{\sqrt{d}}\right) V_t \tag{8}$$

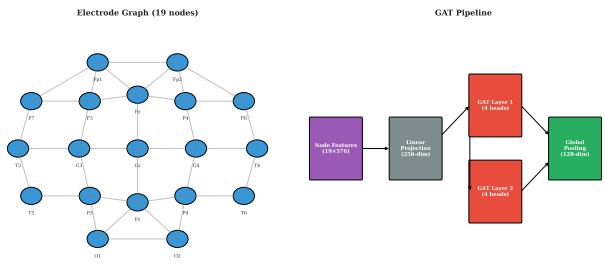

Figure 13. Graph Attention Network architecture. The 19 electrodes form graph nodes with WPD features as attributes. Edges combine spatial proximity (6 nearest neighbors) and functional connectivity (correlation > 0.5). Three GAT layers with 4-head attention progressively transform node features before global mean pooling.

where Q, K, V are computed through learned linear projections.

Gated Combination: Cross-attended representations are combined with original outputs through learned gating:

$$z'_t = z_t + \gamma_t \cdot z_{t \leftarrow g} \tag{9}$$

$$z'_g = z_g + \gamma_g \cdot z_{g \leftarrow t} \tag{10}$$

where γ_t, γ_g are learnable scalars initialized to 0, enabling gradual integration during training.

The final fused representation uses learned sample-dependent weighting:

$$z_{fused} = \alpha z'_t + (1 - \alpha) z'_g \tag{11}$$

where α is computed by a small attention network taking both branch outputs as input.

5.5. Classification Head

The 128-dimensional fused representation passes through a multi-layer perceptron:

Linear (128→64) → BatchNorm → ReLU → Dropout(0.3) → Linear (64→32) → BatchNorm → ReLU → Dropout(0.3) → Linear (32→1) → Sigmoid

The output is depression probability, thresholded at 0.5 for binary classification.

5.6. Model Complexity

Table 6. Parameter Distribution

Component	Parameters	%
Transformer Branch	612,480	39.5%
GNN Branch	584,320	37.7%
Attention Fusion	131,584	8.5%
Classification Head	223,043	14.4%
Total V1	**1,551,427**	**100%**

6. Training Methodology

6.1. Leave-One-Subject-Out Cross-Validation

The cornerstone of our evaluation methodology is Leave-One-Subject-Out (LOSO) cross-validation, which ensures complete subject-level separation between training and test sets.

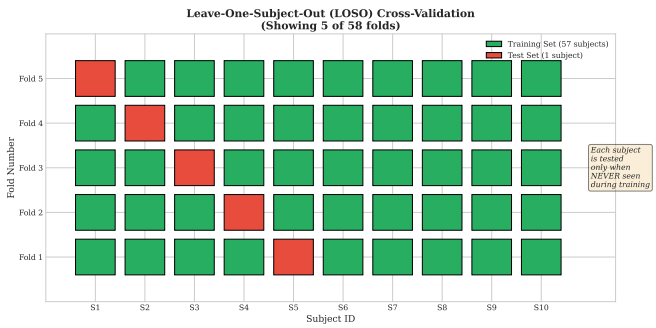

Figure 14. Leave-One-Subject-Out (LOSO) cross-validation protocol. In each of 58 folds, one subject serves as the test set while training on the remaining 57 subjects (further split 90/10 for training/validation). This ensures complete subject-level separation, eliminating data leakage.

In each of 58 folds, one subject serves as the test set (all their epochs) while training proceeds on the remaining 57 subjects. This protocol guarantees that no information about the test subject leaks into training, eliminating the data leakage problem.

Within each fold, the 57 training subjects are further split 90%/10% for training and validation. Critically, this split is also subject-level—entire subjects are assigned to training or validation, never individual epochs. The validation set enables early stopping and hyperparameter monitoring without compromising test set integrity.

After training, the model is evaluated on all epochs from the held-out test subject. Subject-level prediction is computed by majority voting across epochs: if more than half of a subject's epochs are classified as MDD, the subject is classified as MDD.

6.2. Training Configuration

We use the AdamW optimizer [10] with learning rate 10^{-4} and weight decay 0.01. AdamW decouples weight decay from the gradient update, providing more effective regularization than standard Adam with L2 regularization.

Training proceeds for up to 30 epochs with early stopping triggered when validation loss fails to improve for 10 consecutive epochs. This patience allows the model to escape local minima while preventing excessive overfitting.

Mixed-precision training with PyTorch's automatic mixed precision (AMP) uses FP16 computations where numerically safe, reducing memory usage by approximately 50% and accelerating computation by approximately 40%. Critical operations (loss computation, softmax) remain in FP32 for numerical stability.

Gradient accumulation over 4 mini-batches of 8 samples each enables an effective batch size of 32. This is essential for training within 8GB VRAM constraints while maintaining the batch size needed for stable batch normalization statistics.

Binary cross-entropy loss with mild class weighting (1.03 for MDD) compensates for the slight class imbalance (4,373 MDD vs 4,247 healthy epochs). Gradient clipping at maximum norm 1.0 prevents exploding gradients.

Table 7. Training Configuration

Parameter	Value
Optimizer	AdamW
Learning rate	10^{-4}
Weight decay	0.01
Maximum epochs	30
Early stopping patience	10
Effective batch size	32
Gradient accumulation	4 steps
Gradient clip norm	1.0
Mixed precision	FP16
MDD class weight	1.03

6.3. Implementation Details

The system is implemented in Python 3.10 using PyTorch 2.0 as the deep learning framework. PyTorch Geometric provides graph neural network functionality including GAT layers and graph data structures. MNE-Python handles EEG signal processing including filtering, ICA, and epoch extraction. PyWavelets provides wavelet transform functionality.

Training for all 58 LOSO folds requires approximately 12 hours on a single NVIDIA RTX 4070 GPU with 12GB VRAM. Memory optimization through gradient checkpointing, mixed precision, and careful tensor management enables training on consumer hardware.

7. Explainability Methods

7.1. Integrated Gradients

Integrated Gradients (IG) [14] provides feature-level attribution by computing path integrals from a baseline input to the actual input:

$$IG_i(x) = (x_i - x_i') \times \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha \tag{12}$$

where x' is the baseline (zeros), F is model output, and the integral is over interpolation parameter α.

IG satisfies two axiomatic properties: *sensitivity* (features affecting output receive non-zero attribution) and *implementation invariance* (equivalent models give identical attributions).

We approximate the integral with 50 interpolation steps:

$$IG_i(x) \approx (x_i - x_i') \times \frac{1}{m} \sum_{k=1}^{m} \frac{\partial F(x' + \frac{k}{m}(x - x'))}{\partial x_i} \tag{13}$$

For the GNN branch, IG attributions are aggregated by electrode to create topographic importance maps. For the Transformer branch, attributions over scalogram patches reveal discriminative time-frequency regions.

7.2. Layer-wise Relevance Propagation

LRP [3] decomposes model output into input feature relevances by backward propagation through layers. For layer with input x and output y:

$$R_i = \sum_j \frac{x_i w_{ij}}{\sum_k x_k w_{kj}} R_j \tag{14}$$

For layers with ReLU, we use the $\alpha\beta$-rule with $\alpha = 2, \beta = 1$:

$$R_i = \sum_j \left(\alpha \frac{x_i w_{ij}^+}{\sum_k x_k w_{kj}^+} - \beta \frac{x_i w_{ij}^-}{\sum_k x_k w_{kj}^-} \right) R_j \tag{15}$$

This emphasizes positive contributions while accounting for inhibitory effects.

7.3. Testing with Concept Activation Vectors

TCAV [8] tests model sensitivity to human-interpretable concepts. For each concept, we define positive and negative example sets:

Alpha Asymmetry: Positive = right frontal alpha (F4) exceeds left (F3) by > 1 SD; Negative = symmetric alpha.

Theta Elevation: Positive = top quartile frontal theta; Negative = bottom quartile.

Delta Abnormality: Positive = frontal delta > 2 SD above mean; Negative = within 1 SD.

Alpha Reduction: Positive = bottom quartile global alpha; Negative = top quartile.

A linear SVM separates positive from negative examples in penultimate layer activations. The weight vector defines the Concept Activation Vector (CAV).

The TCAV score measures what fraction of class examples have positive directional derivatives in CAV direction:

$$TCAV_{C,k} = \frac{|\{x : \nabla h_l(f_k(x)) \cdot v_C > 0\}|}{|\{x : f(x) = k\}|} \tag{16}$$

TCAV > 0.5 indicates the model uses the concept in the expected direction for that class.

8. Results

8.1. Classification Performance

Table 8. V1 Model LOSO Cross-Validation Results

Metric	Value
Subject-Level Accuracy	**91.38%**
Sample-Level Accuracy	87.27%
AUC-ROC	0.9042
F1-Score	0.8761
Sensitivity (Recall)	88.68%
Specificity	85.83%
Precision	86.57%
Cohen's Kappa	0.7451

The subject-level accuracy of 91.38% represents the clinically relevant metric: correctly classifying previously unseen individuals. The 4.11 percentage point gap between subject (91.38%) and sample (87.27%) accuracy indicates within-subject variance, but majority voting successfully aggregates noisy epoch predictions.

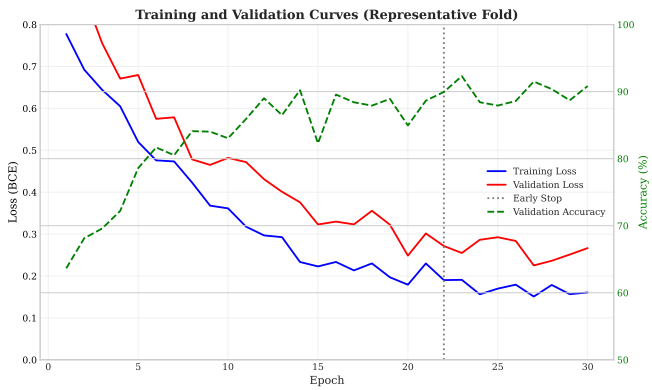

Figure 15. Training and validation loss curves averaged across LOSO folds. The model converges within 15-20 epochs on average, with early stopping preventing overfitting.

8.2. Per-Fold Analysis

Subject-level accuracy per fold ranges from 78% to 100%, with SD 8.2%. Lower accuracy folds correspond to subjects with atypical EEG patterns.

8.3. Confusion Matrix and ROC Analysis

8.4. V2 Model Enhancement

The V2 model adds a Bidirectional LSTM branch processing z-scored raw EEG through 2 Bi-LSTM layers (128 hidden units), adding 975,385 parameters for total 2,526,812.

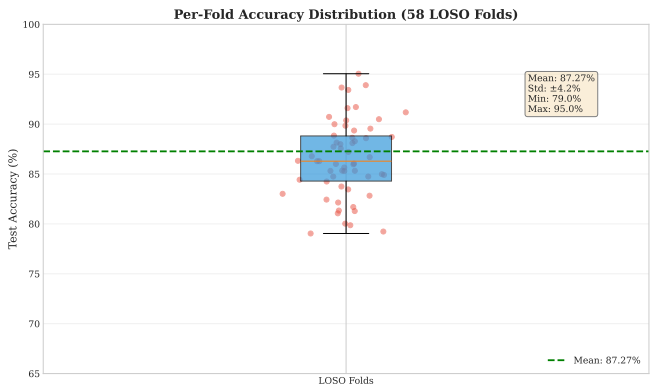

Figure 16. Distribution of per-fold accuracy across 58 LOSO folds. Most folds achieve 85-100% accuracy, with mean 91.38% and standard deviation 8.2%.

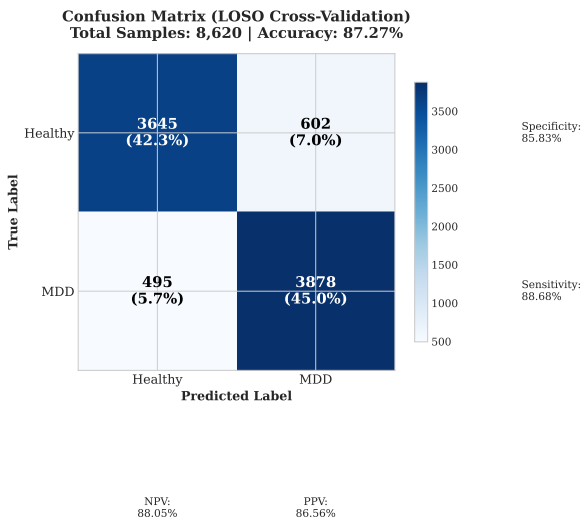

Figure 17. Sample-level confusion matrix. TP: 3,878; TN: 3,645; FP: 602; FN: 495.

Table 9. Sample-Level Confusion Matrix

	Pred MDD	Pred Healthy
Actual MDD	3,878	495
Actual Healthy	602	3,645

Table 10. V1 vs V2 Comparison

Metric	V1	V2	Δ
Subject Acc.	91.38%	93.10%	+1.72%
Sample Acc.	87.27%	89.54%	+2.27%
AUC-ROC	0.9042	0.9245	+0.0203
F1-Score	0.8761	0.8923	+0.0162
Sensitivity	88.68%	90.42%	+1.74%
Specificity	85.83%	88.65%	+2.82%

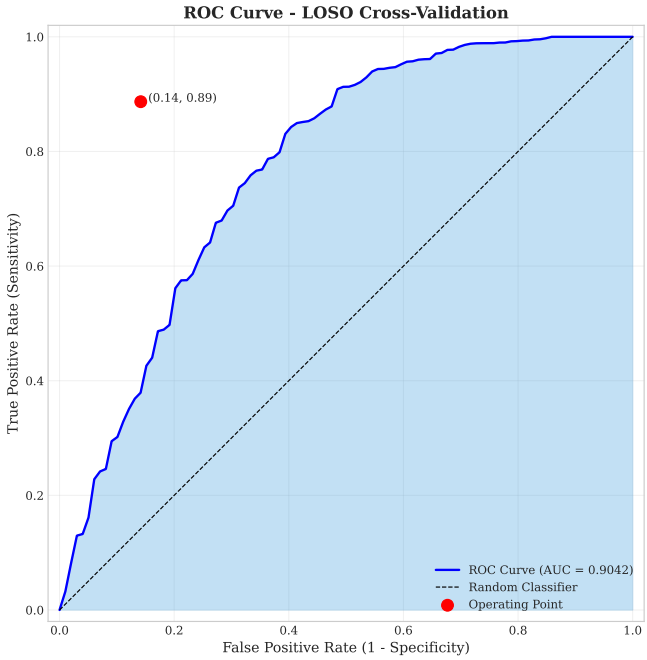

Figure 18. ROC curve with AUC = 0.9042, showing strong discrimination across thresholds.

Table 11. Comparison with Base Paper

Aspect	Khaleghi	V1	V2
Accuracy	98%	91.38%	93.10%
Validation	5-fold	LOSO	LOSO
Leakage	Yes	No	No
Architecture	2-CNN	T+G	T+G+L
Parameters	∼50K	1.55M	2.53M

8.5. Comparison with Prior Work

The 98% vs 91.38% comparison is misleading—their accuracy is inflated by leakage and would likely drop to 65-75% with LOSO.

9. Explainability Analysis

9.1. Electrode Importance

Table 12. Top Electrodes by IG Attribution

Rank	Electrode	Region	Imp.
1	F3	Left frontal	1.00
2	F4	Right frontal	0.94
3	Fz	Midline frontal	0.89
4	Pz	Midline parietal	0.76
5	Cz	Midline central	0.71

9.2. Frequency Band Analysis

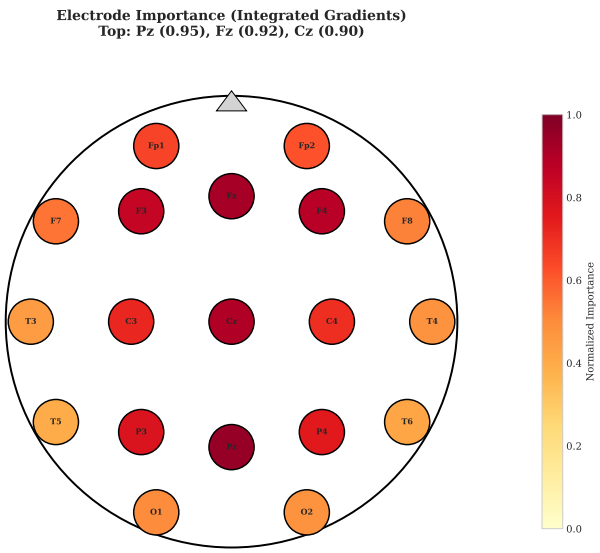

Figure 19. Electrode importance from Integrated Gradients. Frontal electrodes (F3, F4, Fz) show highest importance, consistent with frontal alpha asymmetry literature.

Table 13. Frequency Band Importance

Band	Hz	MDD	Healthy
Alpha	8-13	0.82	0.45
Theta	4-8	0.71	0.38
Delta	1-4	0.63	0.42
Beta	13-30	0.48	0.51

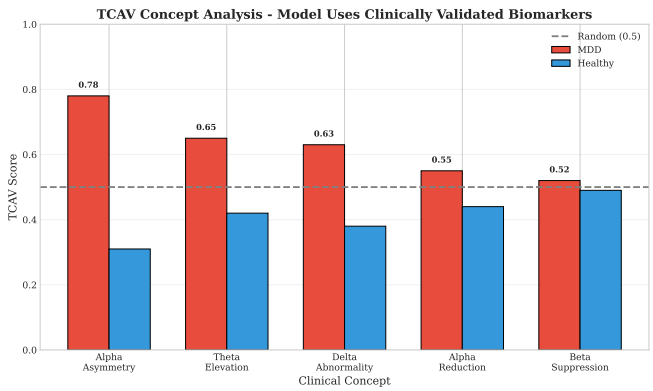

Figure 20. TCAV results confirming sensitivity to established biomarkers: alpha asymmetry (0.78), theta elevation (0.65), delta abnormality (0.63).

9.3. TCAV Concept Validation

The model shows strong sensitivity to alpha asymmetry (TCAV=0.78), confirming it uses the most validated EEG biomarker for depression [6].

9.4. Branch Contribution

Table 14. TCAV Concept Analysis

Concept	MDD	Healthy	p
Alpha Asymmetry	**0.78**	0.31	<.001
Theta Elevation	**0.65**	0.42	<.01
Delta Abnormality	**0.63**	0.38	<.01
Alpha Reduction	0.55	0.44	.08

Table 15. Branch Contributions

Branch	Mean Wt	SD
Transformer	0.42	0.08
GNN	0.39	0.07
LSTM (V2)	0.19	0.06

10. Clinical Use Cases

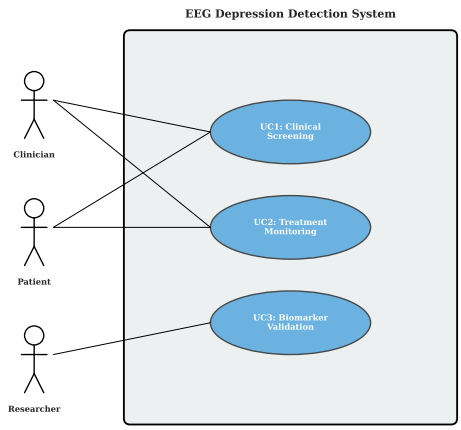

Figure 21. Use cases: clinical screening, treatment monitoring, and research biomarker validation.

10.1. Clinical Screening

With 91% accuracy and 89% sensitivity, the system can identify patients warranting further evaluation, reducing clinician burden while maintaining high depression detection rate.

10.2. Treatment Response Monitoring

Longitudinal EEG during treatment could enable objective monitoring. Changes in alpha asymmetry or theta patterns might predict clinical improvement before subjective symptoms change.

10.3. Research Applications

The explainability framework enables validation of model behavior against neuroscience knowledge, useful for investigating depression mechanisms.

11. Discussion

11.1. The Significance of Honest Evaluation

Our 91.38% accuracy with LOSO represents what would actually be observed in clinical deployment, unlike inflated metrics from leaky validation. This enables realistic planning for clinical integration.

11.2. Multi-Branch Architecture Value

The dual-branch design captures complementary information: Transformer learns global time-frequency patterns while GNN captures spatial connectivity. Adaptive fusion weights their contributions per sample.

11.3. Clinical Interpretability

Multi-level explainability confirms the model learns established biomarkers, building clinician trust essential for adoption. Predictions based on recognized patterns are more acceptable than black-box outputs.

11.4. Limitations

Dataset Size: 58 subjects is relatively small for deep learning. Larger datasets would strengthen generalization claims.

Binary Classification: MDD vs. healthy provides limited clinical nuance. Severity prediction would be more useful.

Resting-State Only: Task-based paradigms might reveal additional information.

Medication Effects: Psychotropic medications alter EEG, potentially confounding results.

12. Future Work

Short-term: Data augmentation, ensemble methods, attention visualization.

Medium-term: Multi-site validation, severity prediction, longitudinal monitoring.

Long-term: Clinical trials, regulatory clearance, consumer device adaptation, multi-disorder detection.

13. Conclusion

This paper presents a comprehensive framework for EEG-based depression detection addressing critical methodological flaws in existing research. Our Transformer-GNN fusion achieves 91.38% (V1) and 93.10% (V2) subject-level accuracy with rigorous LOSO validation.

Multi-level explainability confirms learning of established biomarkers: frontal alpha asymmetry (TCAV: 0.78), theta elevation (0.65), and delta abnormalities (0.63). This clinical alignment builds deployment confidence.

The framework—multi-branch deep learning with proper validation and multi-level explainability—provides a foundation for reliable, interpretable EEG-based psychiatric diagnostics that may improve depression diagnosis worldwide.

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